

The two-dimensional L^p -Mahler conjectures are answered by arXiv:2401.10992

Literature-status note for arXiv:2304.14363

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Status. Explicit later-paper answer to the complete two-dimensional subcase. The higher-dimensional conjectures remain open and imply the classical Mahler conjectures. This note records a literature identification, not a new proof.

1 The source conjectures

Berndtsson, Mastrantonis, and Rubinstein define, for a convex body $K \subset \mathbb{R}^n$ and $p > 0$,

$$h_{p,K}(y) = \frac{1}{p} \log \left(\frac{1}{|K|} \int_K e^{p\langle x, y \rangle} dx \right), \quad \mathcal{M}_p(K) = |K| \int_{\mathbb{R}^n} e^{-h_{p,K}(y)} dy.$$

Their Conjectures 1.3 and 1.4, stated on PDF page 7 of arXiv:2304.14363v1 [1], ask for the following lower bounds.

- For every symmetric convex body $K \subset \mathbb{R}^n$,

$$\mathcal{M}_p([-1, 1]^n) \leq \mathcal{M}_p(K). \tag{1}$$

- For every convex body $K \subset \mathbb{R}^n$,

$$\inf_{x \in \Delta_n} \mathcal{M}_p(\Delta_n - x) \leq \mathcal{M}_p(K), \tag{2}$$

where Δ_n is the standard simplex.

The source formulates these for $p \in (0, \infty]$, with $p = \infty$ recovering the classical symmetric and nonsymmetric Mahler conjectures. It also proves that the family of finite- p conjectures implies the classical conjectures [1, Lemma 3.12].

2 The explicit later answer in dimension two

Mastrantonis and Rubinstein's follow-up *Two-dimensional Blocki, L^p -Mahler, and Bourgain conjectures*, arXiv:2401.10992v3 [2], restates (1) and (2) as its Conjectures 1.1 and 1.2. Its main results on PDF page 3 are exactly:

Theorem 1.3. Conjecture 1.1 holds when $n = 2$.

Theorem 1.4. Conjecture 1.2 holds when $n = 2$.

The paper treats every finite $p > 0$, in both the symmetric and nonsymmetric settings. It also notes immediately afterward that the endpoint $p = \infty$ in dimension two is Mahler’s classical theorem. Consequently the entire $n = 2$ specialization of source Conjectures 1.3–1.4 is resolved for $p \in (0, \infty]$.

The identification is direct rather than inferred: the follow-up cites the source conjectures by number, reproduces their formulas, and labels its Theorems 1.3–1.4 as their two-dimensional verification. The proof uses Mahler sliding of polygon vertices and controls the non-polyhedral L^p -polar along the slide.

3 What remains open

This is a complete answer only in dimension two. No bounded search through 13 August 2026 found a general higher-dimensional resolution. Attempting the unrestricted source target inside this lane would amount to attacking the classical Mahler conjectures, since the source’s finite- p family implies them. The responsible disposition is therefore:

$n = 2$: answered by arXiv:2401.10992; higher dimensions: major open problem.

The local source corpus, exact title/conjecture searches, and later L^p -Mahler papers were checked. The 2024 two-dimensional paper is the precise match; later functional and heat-flow work does not claim the full higher-dimensional inequalities.

References

- [1] B. Berndtsson, V. Mastrantonis, and Y. A. Rubinstein, *L^p -polarity, Mahler volumes, and the isotropic constant*, arXiv:2304.14363v1 (2023).
- [2] V. Mastrantonis and Y. A. Rubinstein, *Two-dimensional Blocki, L^p -Mahler, and Bourgain conjectures*, arXiv:2401.10992v3; to appear in Indiana Univ. Math. J.